ASE 330M HW1 01/26/26

1 a) $f(x) = x^2$

i. $f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)(x-a)^2}{2} \ldots$

$= \lim_{n\to\infty} \sum_{i=0}^{n} \frac{f^{i}(a)(x-a)^{i}}{i!}$

about $x=0 \Rightarrow a=0$.

$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2} + \frac{f^3(0)x^3}{3!} \ldots$

$f(x) = x^2 \rightarrow f(0) = 0$

$f'(x) = 2x \rightarrow f'(0) = 0$

$f''(x) = 2 \rightarrow f''(0) = 2$

$f^3(x) = 0 \rightarrow f^3(0) = 0$

$$\boxed{f(x) = \frac{0x^0}{0!} + \frac{0}{1!}x^1 + \frac{2}{2!}x^2 + \frac{0x^3}{3!} + \ldots \frac{0x^n}{n!} = x^2}$$

after 3 terms, the Taylor approximation is exact.

ii about $x=1 \Rightarrow (a=1)$

$f(x) = f(1) + f'(1)\cdot(x-1) + \frac{f''(1)(x-1)^2}{2} + \frac{f^3(1)(x-1)^3}{3!}$.

$f(1) = 1$

$f'(1) = 2$

$f''(1) = 2$

$f^3(x) = 0$

$f^n(x) = 0$ for $n \geq 3$.

$$\boxed{\begin{aligned} f(x) &= 1 + 2(x-1) + \frac{2(x-1)^2}{2} + \frac{0\cdot(x-1)^3}{3!} + \frac{0(x-1)^4}{4!} \ldots \frac{0(x-1)^n}{n!} \\ &\Downarrow \\ f(x) &= 1 + (2x-2) + (x^2-2x+1) = x^2 \end{aligned}}$$

Still exact after 3 terms of Taylor Series

iii. $f(x)=x^2$ is a 2nd order polynomial so all derivatives of order ≥ 3 are 0. So, after the quadratic term, the Taylor series will stop, leading to a finite series; the different centering points result in different basis polynomials x^k vs. $(x-1)^k$ but both series represent the same function and simplify to x^2 exactly → Taylor's theorem guarentees that the Taylor polynomial will exactly reproduce the function regardless of expansion point, if sufficient derivatives are considered.

1b)

i $f(x)=e^{ax}$ about $x=0$.

$f(0)=e^0=1$

$f'(x)=e^{ax}\cdot a$; $f^n(x)=a^n e^{ax}=a^n f(x)$

$f'(0)=e^0 a=a$; $f^n(0)=a^n$

$$f(x)=f(0)+f'(0)(x-0)+\frac{f''(0)(x-0)^2}{2!}\ldots+\frac{f^n(0)(x-0)^n}{n!}$$

$$\boxed{f(x)=1+ax+\frac{a^2x^2}{2!}+\ldots\frac{a^nx^n}{n!}}$$

$$\boxed{f(x)=\lim_{n\to\infty}\sum_{i=0}^{n}\frac{a^nx^n}{n!}}$$

ii $f(x)=\sin(ax)$ about $x=0$ $\quad f(0)=0$

$f'(x)=a\cos(ax)$ $\quad f'(0)=a$

$f''(x)=-a^2\sin(ax)$ $\quad f''(0)=0$

$f^3(x)=-a^3\cos(ax)$ $\quad f^3(0)=-a^3$

$f^4(x)=a^4\sin(ax)$ $\quad f^4(0)=0$

$$\sin(ax) = \lim_{n\to\infty} \sum_{i=0}^{n} \frac{a^{2n+1}\, x^{2n+1}}{(2n+1)!} (-1)^n = \lim_{n\to\infty} \sum_{i=0}^{n} \frac{(ax)^{2n+1} (-1)^n}{(2n+1)!}$$

iii

$f(x) = \cos(ax)$ $\quad f(0) = 1$

$f'(x) = -a\sin(ax)$ $\quad f'(0) = 0$

$f''(x) = -a^2\cos(ax)$ $\quad f''(0) = -a^2$

$f^3(x) = a^3\sin(ax)$ $\quad f^3(0) = 0$

$f^4(x) = a^4\cos(ax)$ $\quad f^4(0) = a^4$

$$\boxed{\cos(ax) = \lim_{n\to\infty} \sum_{i=0}^{n} \frac{(ax)^{2n} (-1)^n}{2n!}}$$

2 A linear function is a map f from $\mathbb{R}^n \to \mathbb{R}^m$ where

- $f(c_1 x_1 + c_2 x_2) = c_1 f(x_1) + c_2 f(x_2)$ $\quad \forall\, c_1, c_2 \in \mathbb{R}$ &
- $f(0) = 0$. (function passes through origin) $\quad \forall\, x_1, x_2 \in \mathbb{R}^n$

a $f(x_1, x_2) = 3x_1 - x_2 + 1$ $\quad$ let $c_1, c_2 = 1, 1$

$f(a, b) = 3a - b + 1$

where $x_1, x_2 \in \mathcal{X}_1, \mathcal{X}_2$ & $a, b \in \mathcal{X}_1, \mathcal{X}_2$.

$$f(a+x_1, b+x_2) = 3(x_1 + a) - (x_2 + b) + 1$$
$$= 3x_1 - x_2 + 1 + 3a - b$$
$$= f(x_1, x_2) + f(a, b) - 1 \neq f(x_1, x_2) + f(a, b)$$

So, $f(x_1, x_2) = 3x_1 - x_2 + 1$ is a non linear function

alternatively, $f(0,0) = 1$ does not pass through origin.

b $f(x_1, x_2) = x_1 x_2$

$a_1 \in X_1$ & $a_2 \in X_2$, where $x_1 \in X_1$ & $x_2 \in X_2$

$$f(x_1 + a_1, x_2 + a_2) = (x_1 + a_1)(x_2 + a_2) = x_1 x_2 + a_1 a_2 + x_1 a_1 + x_2 a_2$$

$$= f(x_1, x_2) + f(a_1, a_2) + x_1 a_1 + x_2 a_2$$

$$\neq f(x_1, x_2) + f(a_1, a_2)$$

So, $f(x_1, x_2) = x_1 x_2$ is non linear

c $f(g) = \frac{dg(x)}{dx} \rightarrow f(0) = \frac{d0}{dx} = 0$ also:

$$f(g+h) = \frac{d(g(x) + h(x))}{dx} = \frac{dg(x)}{dx} + \frac{dh(x)}{dx}$$

$$= f(g) + f(h)$$

So, $f(g) = \frac{dg(x)}{dx}$ is linear.

d $f(g) = \int_\alpha^\beta g(x)\,dx \rightarrow f(0) = \int_\alpha^\beta 0\,dx = 0$ also:

$$f(ag + bh) = \int_\alpha^\beta ag(x) + bh(x)\,dx \quad , a, b \in \mathbb{R}$$

$a, b \in \mathbb{R}$
$g \in G$ &
$h \in G$

$$= \int_\alpha^\beta ag(x)\,dx + \int_\alpha^\beta bh(x)\,dx$$

$$= a\int_\alpha^\beta g(x)\,dx + b\int_\alpha^\beta h(x)\,dx$$

$$= af(g) + bf(h)$$

So, $f(g) = \int_\alpha^\beta g(x)\,dx$ is linear.

3. a system $\dot{x} = f(x, u)$
is linear if it can be written in the form $\dot{x} = Ax$ for constant matrices A, B or operators
where x is the state.

$\dot{P}_x = v_x$, $\dot{P}_y = v_y$, $\dot{P}_z = v_z$ are linear

but,
1) $\dot{v}_x = g\tan\theta$, 2) $\dot{v}_y = -g\tan(\phi)$, 3) $\dot{v}_z = \tau - g$

are non linear systems since:

g is a scalar constant, so f is not parameterized by g

1) $f(\theta) = g\tan\theta$ is a non-linear function

$$f(\theta_1+\theta_2) = g\tan(\theta_1+\theta_2) \neq g\tan\theta_1 + g\tan\theta_2$$
$$= g(\tan\theta_1 + \tan\theta_2)$$
$$= f(\theta_1) + f(\theta_2)$$

for example, let $\theta_1 = 30^\circ$ & $\theta_2 = 30^\circ$

$\tan(\theta_1+\theta_2) = \tan(60^\circ) = \frac{\sqrt{3}\ 2}{2 \cdot 1} = \sqrt{3}$

$\tan(\theta_1) + \tan(\theta_2) = 2 \cdot \tan(30^\circ)$

$= 2\ \frac{1/2}{\sqrt{3}} \cdot 2 = \frac{1}{\sqrt{3}}$

2) $f(\phi) = -g\tan\phi$ is also non-linear by the same argument in 1). $f(\phi_1+\phi_2) = -g\tan(\phi_1+\phi_2)$
$\neq g[\tan\phi_1 + \tan\phi_2]$
$= f(\phi_1) + f(\phi_2)$

3) $\dot{v}_z = \tau - g$

$f(\tau) = \tau - g$; $f(\tau=0) = 0 - g = -g \neq 0$.

so, $f(\tau)$ is non-linear.

Variables θ, ϕ, and τ make the system non-linear.

With a small angle approximation,

$\tan(\theta) \approx \theta$, $\tan(\phi) \approx \phi$

so, we can get a linear affine model (still not linear!)

(only the $\dot{v}_z = \tau - g$ non-linearity)

b)

$$\frac{d}{dt}\begin{bmatrix} p_x \\ p_y \\ p_z \\ v_x \\ v_y \\ v_z \end{bmatrix} = \begin{bmatrix} v_x \\ v_y \\ v_z \\ g\tan\theta \\ -g\tan\phi \\ \tau - g \end{bmatrix}$$

define $\tilde{\theta} = \tan\theta$, $\tilde{\phi} = \tan\phi$, & $\tilde{\tau} = \tau - g$

which is valid when $\theta, \phi \in (-\pi/2, \pi/2)$.

then:

$$\frac{d}{dt}\begin{bmatrix} p_x \\ p_y \\ p_z \\ v_x \\ v_y \\ v_z \end{bmatrix} = \begin{bmatrix} v_x \\ v_y \\ v_z \\ g\tilde{\theta} \\ -g\tilde{\phi} \\ \tilde{\tau} \end{bmatrix}$$

let $x = \begin{bmatrix} p_x \\ p_y \\ p_z \\ v_x \\ v_y \\ v_z \end{bmatrix}$ then we can write the system of equations in $\dot{x} = Ax + B\underset{\sim}{u}$

inputs to system

$u = \begin{bmatrix} \tilde{\theta} \\ \tilde{\phi} \\ \tilde{\tau} \end{bmatrix}$

$$\dot{x} = \frac{d}{dt}\underbrace{\begin{bmatrix} p_x \\ p_y \\ p_z \\ v_x \\ v_y \\ v_z \end{bmatrix}}_{x} = \overbrace{\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}}^{A} \overbrace{\begin{bmatrix} p_x \\ p_y \\ p_z \\ v_x \\ v_y \\ v_z \end{bmatrix}}^{x} + \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ g & 0 & 0 \\ 0 & -g & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{B} \underbrace{\begin{bmatrix} \tilde{\theta} \\ \tilde{\phi} \\ \tilde{\tau} \end{bmatrix}}_{u}$$

The coordinate change eliminates the non-linear functions (tan) and replaces them with additional inputs $\vec{u}$.

4. $\ddot{\theta}_1 = -2\theta_1 + \theta_2$

$\ddot{\theta}_2 = 2\theta_1 - 2\theta_2$

$$\frac{d}{dt}\begin{bmatrix} \theta_1 \\ \theta_2 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & 1 & 0 & 0 \\ 2 & -2 & 0 & 0 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

a) state variables: $\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2$

4b)

$$\frac{d}{dt}\underbrace{\begin{bmatrix}\theta_1\\ \theta_2\\ \dot\theta_1\\ \dot\theta_2\end{bmatrix}}_{x} = \underbrace{\begin{bmatrix}0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\\ -2 & 1 & 0 & 0\\ 2 & -2 & 0 & 0\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}\theta_1\\ \theta_2\\ \dot\theta_1\\ \dot\theta_2\end{bmatrix}}_{x}$$

$$\frac{d}{dt}x = A\,x$$

5. $m\ddot{z} + d\dot{z} + kz = 0 \rightarrow \ddot{z} = -\frac{d}{m}\dot{z} - \frac{k}{m}z$

a) State variables: $z, \dot{z}$

b) $\dot{x} = Ax$ format:

$$\frac{d}{dt}\underbrace{\begin{bmatrix}z\\ \dot z\end{bmatrix}}_{\vec{z}} = \begin{bmatrix}\dot z\\ \ddot z\end{bmatrix} = \underbrace{\begin{bmatrix}0 & 1\\ -\frac{k}{m} & -\frac{d}{m}\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}z\\ \dot z\end{bmatrix}}_{\vec{z}}$$

6.

a) $\ddot{z} = -z$; state variable: $z, \dot{z}$

$$\frac{d}{dt}\underbrace{\begin{bmatrix}z\\ \dot z\end{bmatrix}}_{\vec{z}} = \underbrace{\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}z\\ \dot z\end{bmatrix}}_{\vec{z}}$$

$$\frac{d}{dt}\vec{z} = A\,\vec{z}$$

Alternate: any linear invertible transform of $(z, \dot{z})$ also works. e.g:

$$\frac{d}{dt}\begin{bmatrix}z+\dot z\\ \dot z\end{bmatrix} = \begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\begin{bmatrix}z\\ \dot z\end{bmatrix}$$

b) $\ddot{z} - \dot{z} + 2z = 0$ $\qquad$ $\ddot{z} = \dot{z} - 2z$

$$\frac{d}{dt}\underbrace{\begin{bmatrix} z \\ \dot{z} \end{bmatrix}}_{\vec{z}} = \underbrace{\begin{bmatrix} 0 & 1 \\ -2 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} z \\ \dot{z} \end{bmatrix}}_{\vec{z}}$$

$$\frac{d}{dt}\vec{z} = A\vec{z}$$

state variables $z, \dot{z}$

state variables: $z, \dot{z}$

(again, any linear invertible transform like $\frac{d}{dt}\begin{bmatrix} az \\ \dot{z} \end{bmatrix} = \begin{bmatrix} 0 & a \\ -\frac{2}{a} & 1 \end{bmatrix}\begin{bmatrix} az \\ \dot{z} \end{bmatrix}$ also works)

c) $\ddot{z} + z = e^{2t}$

$$\frac{d}{dt}\underbrace{\begin{bmatrix} z \\ \dot{z} \\ e^{2t} \end{bmatrix}}_{\vec{z}} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} z \\ \dot{z} \\ e^{2t} \end{bmatrix}}_{\vec{z}}$$

$$\frac{d}{dt}\vec{z} = A\vec{z}$$

state variables $z, \dot{z}, e^{2t}$

other option: treat e^{2t} as a linear-time-varying input: (affine) (not-self-contained).

$$\frac{d}{dt}\begin{bmatrix} z \\ \dot{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} z \\ \dot{z} \end{bmatrix} + \begin{bmatrix} 0 \\ \underbrace{u(t)}_{e^{2t}} \end{bmatrix}$$

d) $\ddot{z} = \cos(t)$

$$\frac{d}{dt}\begin{bmatrix} z \\ \dot{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} z \\ \dot{z} \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\cos t$$

Time dependent system

for linearity, instead, use:

$$\frac{d}{dt}\underbrace{\begin{bmatrix} z \\ \dot{z} \\ \cos t \\ -\sin t \end{bmatrix}}_{\vec{z}} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}}_{A}\underbrace{\begin{bmatrix} z \\ \dot{z} \\ \cos t \\ -\sin t \end{bmatrix}}_{z}$$

state variables: $z, \dot{z}, \cos t, \sin t$

$$\frac{d}{dt}\vec{z} = A\, z$$

With <u>fewer variables</u>, the linear system can be written as: (state space: $\dot{z}$, $\cos t$, $\sin t$).

$$\frac{d}{dt}\underbrace{\begin{bmatrix} \dot{z} \\ \cos t \\ \sin t \end{bmatrix}}_{\vec{z}} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}}_{A}\underbrace{\begin{bmatrix} \dot{z} \\ \cos t \\ \sin t \end{bmatrix}}_{\vec{z}}$$

$$\frac{d}{dt}\vec{z} = A\,\vec{z}$$

another option would be to multiply the state space by any constant $c \in \mathbb{R}$.